PhenoCam vs model transition dates, PhenoCam BDT pixels, 30-d nowcast windows anchored on the fixed-DOY-band event window (ev2)

LCSIF window (observed)

TFT

PhenoFormerSeq

4GST (OLS extrap.)

n = 507 site-yr, bias -4 d, MAE 11 d

n = 507 site-yr, bias -5 d, MAE 11 d

n = 506 site-yr, bias -6 d, MAE 12 d

n = 76 site-yr, bias +8 d, MAE 17 d
85% of forecasts never cross

SOS: model date (DOY)

PhenoCam date (DOY)

within-site r = 0.37
49 sites ≥5 y, slope 0.11

within-site r = -0.09
49 sites ≥5 y, slope -0.04

within-site r = -0.05
49 sites ≥5 y, slope -0.02

no site with ≥5 matched years
(too many forecasts never cross)

SOS: model anomaly (d)

PhenoCam anomaly (d)

n = 547 site-yr, bias -11 d, MAE 16 d

n = 547 site-yr, bias -11 d, MAE 16 d

n = 547 site-yr, bias -10 d, MAE 16 d

n = 290 site-yr, bias -3 d, MAE 15 d
47% of forecasts never cross

EOS: model date (DOY)

PhenoCam date (DOY)

within-site r = -0.10
56 sites ≥5 y, slope -0.01

within-site r = -0.17
56 sites ≥5 y, slope -0.03

within-site r = -0.16
56 sites ≥5 y, slope -0.02

within-site r = -0.11
21 sites ≥5 y, slope -0.03

EOS: model anomaly (d)

PhenoCam anomaly (d)

test period 2008–10

train period ≥2011 (in-sample)

2011–2021 lie in the training years of the TFT and PhenoFormerSeq checkpoints (in-sample); only 2008–2010 are test years. LSTM omitted (categorical embeddings not saved).